

Overview of Batch normalization

Subject: Batch normalization

$$\hat{X}_i = \frac{X_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

$$y_i = \gamma \hat{X}_i + \beta$$

1.

$f_{BN}(w, \gamma) = E_x [\phi(\gamma x^T w (w^T S w)^{-1/2})]$ $\{\displaystyle f_{BN}(w, \gamma) = E_x [\phi(x^T w / \sqrt{w^T S w})]\}$, where $(w^T S w)^{-1/2}$ $\{\displaystyle (w^T S w)^{-1/2}\}$ is the induced norm of S $\{\displaystyle S\}$, $\|w\|_S$ $\{\displaystyle \|w\|_S\}$. Hence, it could be concluded that $f_{BN}(w, \gamma) = E_x [\phi(x^T \tilde{w})]$ $\{\displaystyle f_{BN}(w, \gamma) = E_x [\phi(x^T \tilde{w})]\}$, where $\tilde{w} = \gamma w / \|w\|_S$ $\{\displaystyle \tilde{w} = \gamma w / \|w\|_S\}$, and γ $\{\displaystyle \gamma\}$ and w $\{\displaystyle w\}$ account for its length and direction separately. This property could then be used to prove the faster convergence of problems with batch normalization.

2.

$h(w_t, \gamma_t) = E_z [\phi'(z^T \tilde{w}_t)] (u^T w_t) - E_z [\phi''(z^T \tilde{w}_t)] (u^T w_t)^2$ $\{\displaystyle h(w_t, \gamma_t) = E_z [\phi'(z^T \tilde{w}_t)] (u^T w_t) - E_z [\phi''(z^T \tilde{w}_t)] (u^T w_t)^2]\}$. Let the step size be

3.

Measure Since it is hypothesized that batch normalization layers could reduce internal covariate shift, an experiment is set up to measure quantitatively how much covariate shift is reduced. First, the notion of internal covariate shift needs to be defined mathematically. Specifically, to quantify the adjustment that a layer's parameters make in response to updates in previous layers, the correlation between the gradients of the loss before and after all previous layers are updated is measured, since gradients could capture the shifts from the first-order training method. If the shift introduced by the changes in previous layers is small, then the correlation between the gradients would be close to 1. The correlation between the gradients are computed for four models: a standard VGG network, a VGG network with batch normalization layers, a 25-layer deep linear network (DLN) trained with full-batch gradient descent, and a DLN network with batch normalization layers. Interestingly, it is shown that the standard VGG and DLN models both have higher correlations of gradients compared with their counterparts, indicating that the additional batch normalization layers are not reducing internal covariate shift.

4.

$\min_{\mathbf{w} \in \mathbb{R}^d \setminus \{0\}} \rho(\mathbf{w}) = \min_{\mathbf{w} \in \mathbb{R}^d \setminus \{0\}} (-\mathbf{w}^T \mathbf{u} + \mathbf{w}^T \mathbf{S} \mathbf{w})$ $\{\displaystyle \min_{\mathbf{w} \in \mathbb{R}^d \setminus \{0\}} \rho(\mathbf{w}) = \min_{\mathbf{w} \in \mathbb{R}^d \setminus \{0\}} (-\mathbf{w}^T \mathbf{u} + \mathbf{w}^T \mathbf{S} \mathbf{w})\}$. Note that this objective is a form of the generalized Rayleigh quotient

5.

Smoothness One alternative explanation is that the improvement with batch normalization is instead due to producing a smoother parameter space and smoother gradients, as formalized by a smaller Lipschitz constant. Consider two identical networks, one contains batch normalization layers and the other does not, the behaviors of these two networks are then compared. Denote the loss functions as $\hat{L}$ $\{\displaystyle \hat{L}\}$ and L $\{\displaystyle L\}$, respectively. Let the input to both networks be x $\{\displaystyle x\}$, and the output be y $\{\displaystyle y\}$, for which $y = Wx$ $\{\displaystyle y=Wx\}$, where W $\{\displaystyle W\}$ is the layer weights. For the second network, y $\{\displaystyle y\}$ additionally goes through a batch normalization layer. Denote the normalized activation as $\hat{y}$ $\{\displaystyle \hat{y}\}$, which has zero mean and unit variance. Let the transformed activation be $z = \gamma \hat{y} + \beta$ $\{\displaystyle z=\gamma \hat{y} + \beta\}$, and suppose γ $\{\displaystyle \gamma\}$ and β $\{\displaystyle \beta\}$ are constants. Finally, denote the standard deviation over a mini-batch $y_j \in \mathbb{R}^m$ $\{\displaystyle \hat{y}_{\{j\}} \in \mathbb{R}^m\}$ as σ_j $\{\displaystyle \sigma_{\{j\}}\}$. First, it can be shown that the gradient magnitude of a batch normalized network, $\|\nabla_{\hat{y}_{\{i\}}} \hat{L}\|$ $\{\displaystyle \|\nabla_{\hat{y}_{\{i\}}} \hat{L}\|\}$, is bounded, with the bound expressed as

6.

$E[x^{(k)}] = E_B[\mu_B^{(k)}]$ $\{\displaystyle E[x^{(k)}] = E_B[\mu_B^{(k)}]\}$, and $\text{Var}[x^{(k)}] = \frac{m}{m-1} E_B[(\sigma_B^{(k)})^2]$ $\{\displaystyle \text{Var}[x^{(k)}] = \frac{m}{m-1} E_B[(\sigma_B^{(k)})^2]\}$. The population statistics thus is a complete representation of the mini-batches. The BN transform in the inference step thus becomes

7.

$\|\tilde{\mathbf{w}} \sim f(\mathbf{w} \sim T_d)\| \leq (1 - \mu_L)^2 T_d \Phi^2(\rho(\mathbf{w}_0) - \rho^*) + 2 - T_s \zeta |b_t(0) - a_t(0)| \mu^2$ $\{\displaystyle \|\tilde{\mathbf{w}} \sim f(\mathbf{w} \sim T_d)\| \leq (1 - \mu_L)^2 T_d \Phi^2(\rho(\mathbf{w}_0) - \rho^*) + 2 - T_s \zeta |b_t(0) - a_t(0)| \mu^2\}$, such that the algorithm is guaranteed to converge linearly. Although the proof stands on the assumption of Gaussian input, it is also shown in experiments that GDNP could accelerate optimization without this constraint.

8.

$(\partial_{\gamma} f_{LH}(\mathbf{w}_t, a_t(T_s)))^2 \leq 2 - T_s \zeta |b_t(0) - a_t(0)| \mu^2$ $\{\displaystyle (\partial_{\gamma} f_{LH}(\mathbf{w}_t, a_t(T_s)))^2 \leq 2 - T_s \zeta |b_t(0) - a_t(0)| \mu^2\}$, where $a_t(0)$ $\{\displaystyle a_t(0)\}$ and $b_t(0)$ $\{\displaystyle b_t(0)\}$ are the two starting points of the bisection algorithm on the left and on the right, correspondingly. Further, for each iteration, the norm of the gradient of f_{LH} $\{\displaystyle f_{LH}\}$ with respect to $\mathbf{w}$ $\{\displaystyle \mathbf{w}\}$ converges linearly, such that

9.

$w_{t+1} = w_t - s_t \nabla f(w_t, \gamma_t)$ $\{\displaystyle w_{t+1}=w_t-s_t\triangledown_{w}f(w_t,\gamma_t)\}$. Then update the length according to